

PAUL DES BROSSES

CREATIVE TECHNOLOGY ENGINEER — INTERACTIVE SYSTEMS & XR

Real-Time Interaction × 3D / XR × Sensors | **Final-Year Internship (6 months) — Feb. 2027**

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Portfolio: paul-des-brosses.github.io/en

EDUCATION

ESILV — Paris 2022 – 2027

M.Eng. in Creative Technology (M2)

Real-time 3D, human-computer interaction (HCI), embedded hardware, applied AI, signal processing.

Erasmus semester at UVIC (Spain, taught in English).

SKILLS

Real-Time 3D & Interaction

Unity 6 (C#, VFX Graph, WebGL) · MediaPipe (real-time tracking) · interactive audio · immersive systems · visual creation & editing (DaVinci Resolve, Photopea/Photoshop)

Code & Architecture

Python, C#, C++ · Linux (embedded & server) · SQL · Git, Docker

Hardware, Sensors & Prototyping

Sensor integration · ESP32, Arduino, embedded C, I2C bus · SolidWorks (CAD), KiCad (PCB) · networking: TCP, UDP, HTTP, MQTT

Applied AI & LLM

LangGraph, LangChain · multi-agent systems · LLM API integration · RAG, local LLMs (Ollama) · Pydantic · ChromaDB

Simulation & Data

Digital twins · system modeling · NumPy, Pandas, Matplotlib

SOFT SKILLS

Project management · autonomy · design thinking · research methodology · perseverance · technical communication

LANGUAGES

French (Native) · English (B2) · Spanish (B1)

INTERESTS

Maker / DIY

PCB design, 3D printing, hardware prototyping

Sports

Swimming, mountain hiking

PROFESSIONAL EXPERIENCE

R&D Engineer (Intern then Freelance) — Ardanti Défense Apr. 2026 – Present

Critical infrastructure & sensitive site security (continued as freelance alongside M2)

- **Intrusion detection meta-sensor** (3D LiDAR + thermal camera fusion on Linux): component selection, software development, network architecture, and real-time reliability.
- Led the project through to **client demonstration** and deployment on a critical site; extended into a **freelance contract** to develop new detection modules.
- Designed a **sovereign local AI system** (RAG, self-hosted LLMs via Ollama, LangGraph) compliant with ANSSI security guidelines.
- Cross-functional contribution to **technical and visual communication** (video, posters) for an industry trade show.

ENGINEERING RESEARCH & PROJECTS

Forest of Senses — Immersion & Implicit Motor Adaptation 2025 – 2026

M1 Research & Innovation Project (HCI) · Unity 6, C#, Python

- Immersive audio-visual environment inducing **implicit hand motor adaptation without instructions**, driven by a 5-finger medical isometric force sensor (*Dextrain*).
- Real-time sensor interfacing in Unity 6, **interactive sound modulation** engine, experimental study with significant results, Python statistical analysis, and research manuscript.

Echoes of a Gesture — Contactless Full-Body Interaction 2026 – Present

M2 Research & Innovation Project (HCI) · Unity VFX Graph, MediaPipe Pose, C#

- M2 research on controller-free full-body interaction: continuous modulation of a generative particle field via the 4 **Laban Effort factors** (Space, Time, Weight, Flow).
- Real-time system architecture design (**MediaPipe 33 keypoints** at 1080p, sliding-window kinematic descriptors) and *within-subject* experimental protocol.

Bocage Digital Twin — Interactive Ecological Digital Twin 2026

Real-time complex system simulation · Unity 6 WebGL, C# (Live online demo)

- Interactive **digital twin** modeling the real-time coupled effects of agricultural choices and climate on water, soil, biodiversity, and yield.
- **Dual-layer simulation** (live run vs. frozen baseline) grounded in public scientific literature.

Applied Fox — Autonomous Local Multi-Agent System 2026

Applied AI — local LLM agents · Python, LangGraph, Ollama, Pydantic

- Collaborative **multi-agent pipeline** (search, integration, arbitration, reporting) for prioritized R&D intelligence with strict *human-in-the-loop* validation, running **100% locally**.

Lightning TDOA Simulator — Compute Engine & Simulation 2025

Algorithmic simulator · Python (Interactive online demo)

- 3-station VLF lightning localization simulator via hyperbolic multilateration (**TDOA**): noise modeling, solver comparison, and Monte Carlo validation.

LEADERSHIP & VOLUNTEERING

Patrol Leader — Scouting 2012 – 2023

Saint Louis Group, Paris (Scouts d'Europe)

- Led a team of 6 youths over several years, managing camp logistics, budget, and autonomous field decisions (11 years of continuous commitment).

Humanitarian Mission in Armenia July 2022

L'Œuvre d'Orient

- Renovated an orphanage (masonry, painting) and led daily activities for children.